MANUZAM

A PATTERN-BASED AND MULTI-SIGNAL APPROACH FOR FAILURE PREDICTION IN THE MANUFACTURING INDUSTRY

CYBER PHYSICAL & SOCIAL SYSTEMS

PRESENTED BY M ZAIN UL ABIDEEN

OVERVIEW

MANUZAM IS A CYBER-PHYSICAL SYSTEM (CPS) THAT SUPPORTS PREDICTIVE MAINTENANCE IN INDUSTRIAL ENVIRONMENTS BY IDENTIFYING EARLY MACHINE FAILURE INDICATORS THROUGH SOUND AND SENSOR DATA ANALYSIS. THINK OF IT AS SHAZAM FOR MACHINERY. USERS CAN RECORD OR UPLOAD MACHINE AUDIO SAMPLES, LABEL THEM (E.G., NORMAL, FAULTY), AND TRAIN CUSTOM MODELS TO DETECT ANOMALIES. THE SYSTEM ALSO INCORPORATES TEMPERATURE AND VIBRATION SENSOR DATA FOR ENHANCED, CONTEXT-AWARE FAULT DETECTION.

KEY FEATURES

- USER-DRIVEN MACHINE AUDIO DATASET
- MANAGEMENT OF MULTIPLE MACHINE PROFILE
- MODEL TRAINING AND TESTING BY USERS
- REAL TIME FAULT DETECTION
- SENSOR IMPLEMENTATION FOR MORE PRECISE PREDICTION

Use Cases

SINGLE USER ROLES

Model Training

USER CAN UPLOAD TRAINING AND TEST AUDIO

Fault Recognition

THE USER CAN LABEL THE DETECTED FAULT FOR FUTURE PERFORMANCE

Performance Dashboard

VIEW PAST RESULTS AND MANAGE THEIR DATASET

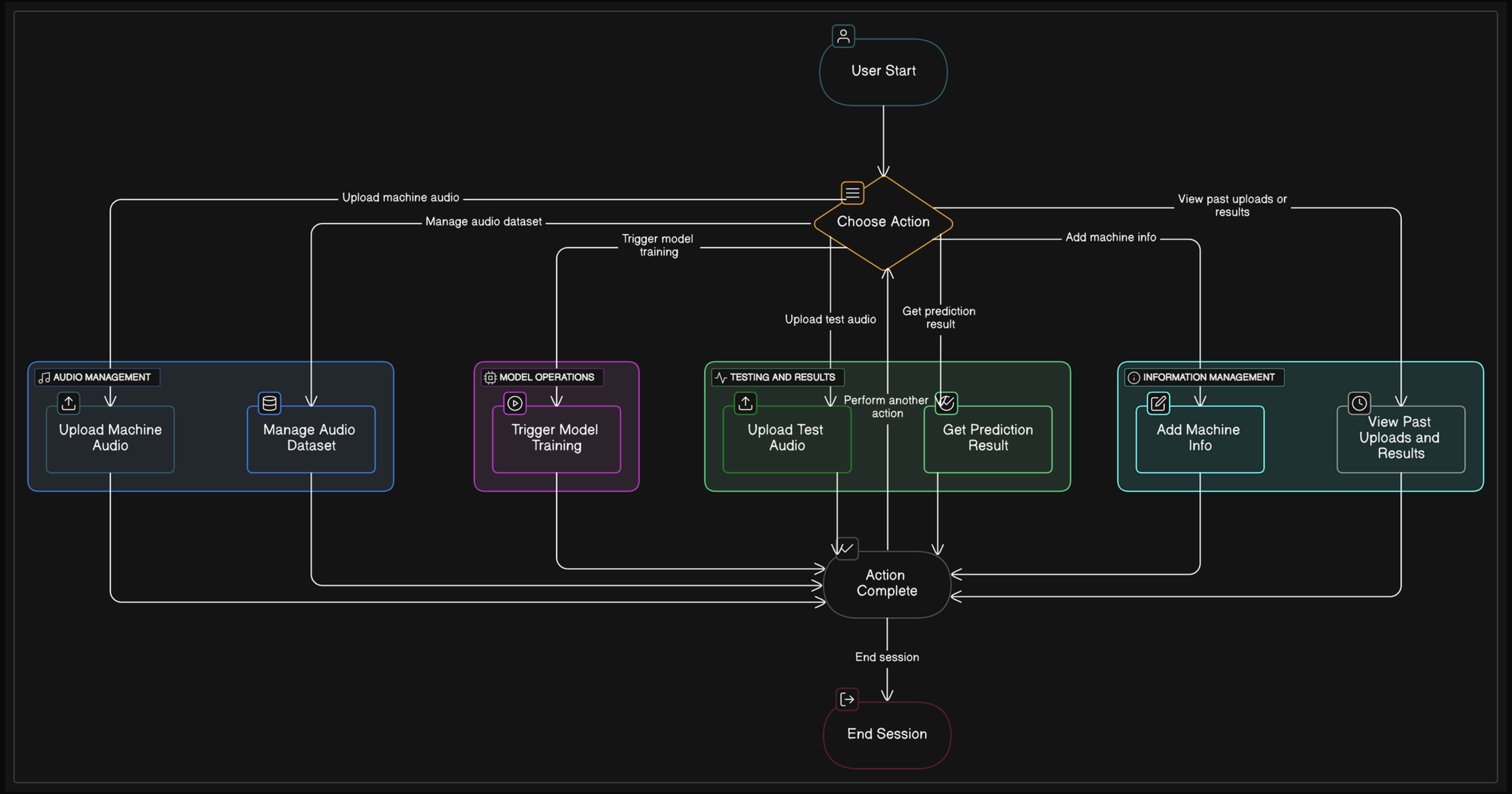
Profile management

MANAGE MULTIPLE, PERSONALIZED MACHINE PROFILES

Real-time testing:

TRIGGER MODEL TRAINING AND TESTING

USE CASE DIAGRAM



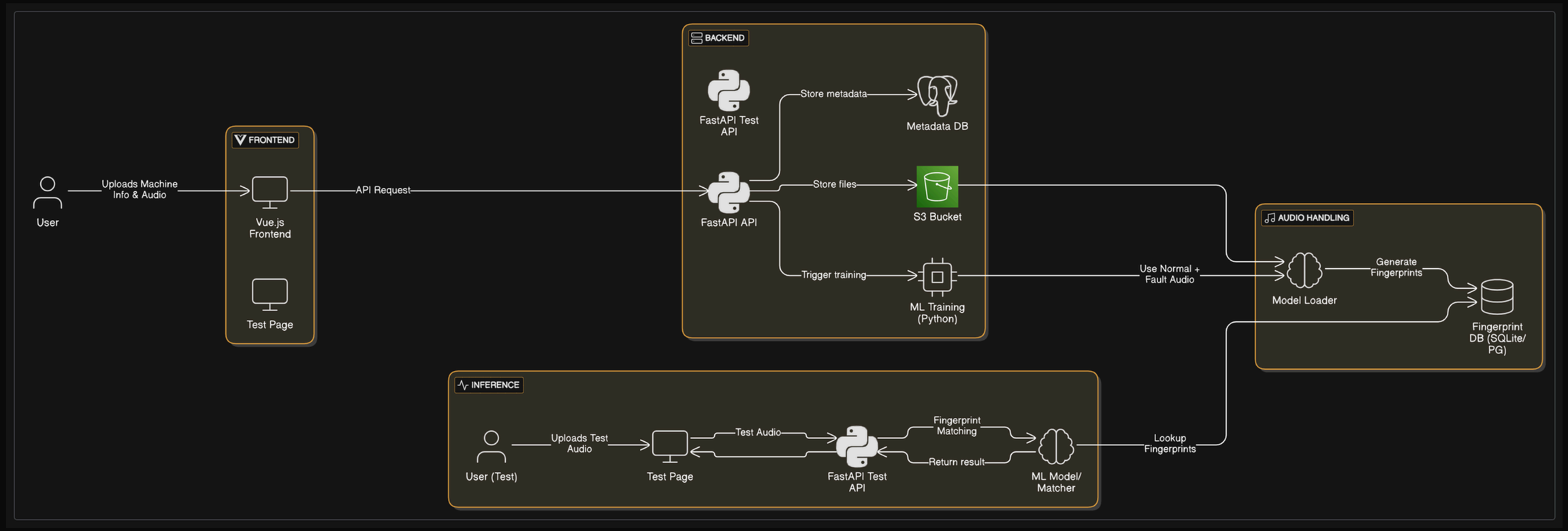
TECHNICAL COMPONENTS

- FRONTEND: VUE JS WITH CHART JS
- BACKEND: FAST API (PYTHON)
- DATABASE: SQLITE
- CLOUD STORAGE: S3 BUCKET
- ML LIBRARY: SCIKIT-LEARN

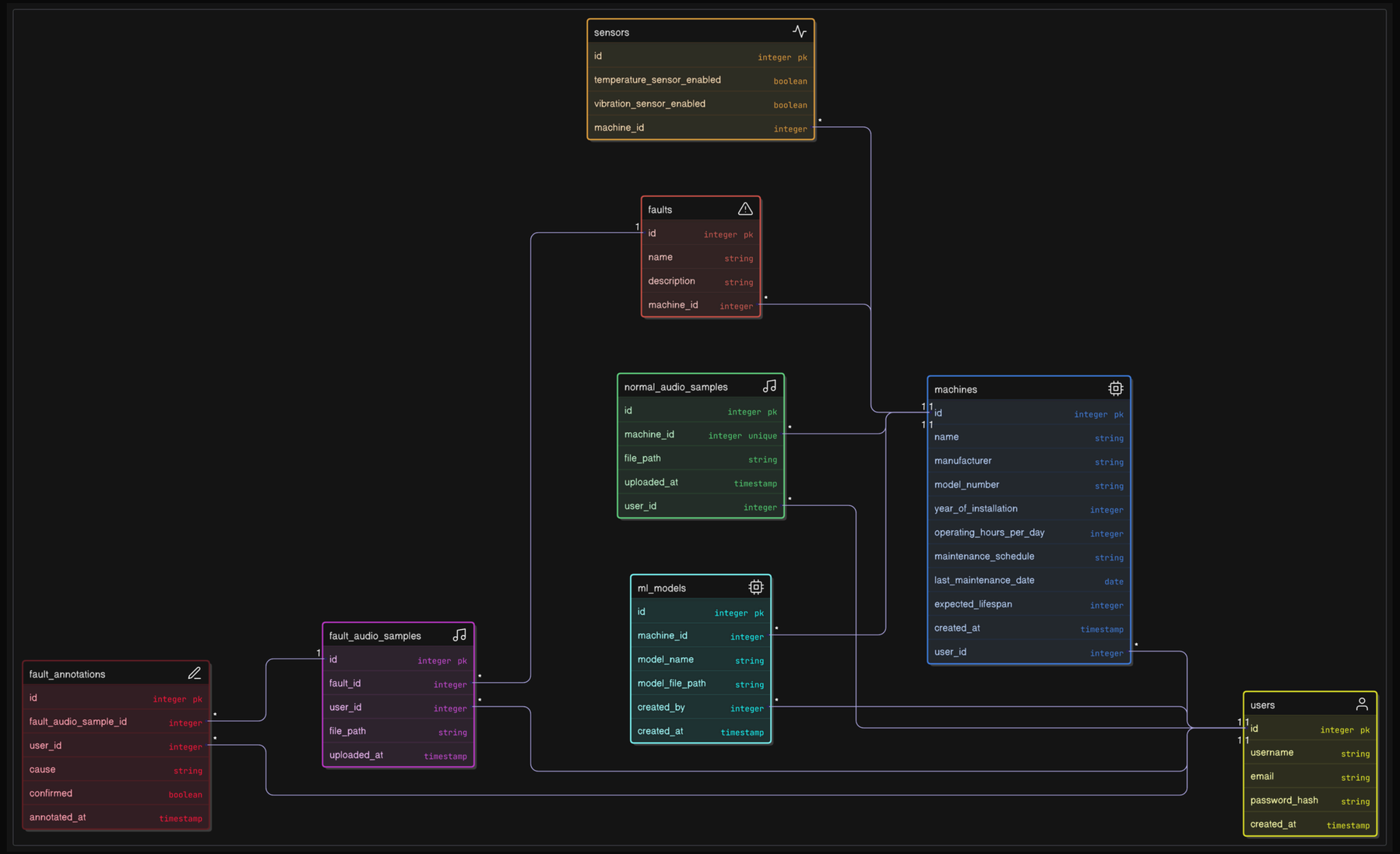
SYSTEM ARCHITECTURE

- FRONTEND FOR UPLOADING AUDIO, VIEWING RESULTS
- BACKEND TO HANDLE MODEL TRAINING, APIS, STORAGE, FINGERPRINTING, AND PREDICTION
- SQLITE DATABASE TO STORE FINGERPRINTS AND USER-SPECIFIC MACHINE DATA
- PYTHON BACKEND FOR AUDIO PROCESSING AND MODEL LOGIC
- S3 BUCKET FOR AUDIO STORAGE

SYSTEM ARCHITECTURE DIAGRAM



DATABASE SCHEMA DESIGN



FIGMA DESIGN

SUMMARY

THIS MANUZAM SYSTEM ENABLES USERS TO DETECT MACHINE FAULTS BY ANALYZING AUDIO SIGNATURES.

IT SUPPORTS:

- PERSONALIZED MACHINE PROFILES
- END-TO-END AUDIO PROCESSING
- USER-MANAGED DATASETS AND RESULTS
- POTENTIAL INTEGRATION WITH IOT SENSORS AND REAL-TIME SYSTEMS IN FUTURE.

THANK YOU FOR YOUR ATTENTION!

PLEASE FEEL FREE TO ASK ANY QUESTIONS OR OFFER SUGGESTIONS.